Reproducing “Vision Transformers Need Registers”

Final technical report — Machine Learning paper reproduction and deployment

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Code, results and notebooks:
https://github.com/recommended131225435/vit-registers-reproduction

Branch `main`: reproduction and deployment · branch `experiment-model-size`: the additional experiment

Abstract. Darcet et al. (ICLR 2024) found that large Vision Transformers quietly turn a few background patches into storage for image-wide information, which shows up as bright, meaningless spots in their attention maps, and that giving the model a handful of extra “register” tokens removes the problem. We reproduce this in two ways. First, on Meta’s released DINOv2 models, we measure the high-norm tokens directly: 2.62% of ViT-g patch tokens exceed the paper’s cutoff (paper: 2.37%), they sit on redundant background patches, they vanish with registers, and the large values move into the registers. Second, we implement a ViT with registers from scratch and train it on Tiny-ImageNet (38.99% vs 37.86% validation accuracy without/with registers). Our own experiment trains ViTs of increasing size and measures whether the artifacts appear. Across ViT-Tiny, ViT-Small and ViT-Base (5M–85M parameters, each trained for 50 epochs with and without 4 registers) we find no high-norm tokens at all: the largest token in the last block is at most 2.64× the median, and the registers receive only the attention an average token would get (5.7–6.2% against 5.8%). Registers change accuracy by less than one point (−0.81, +0.44, −0.66). Together with the DINOv2 results, where clear outliers appear only in the 1.1B-parameter ViT-g, this supports the paper’s view that artifacts, and the need for registers, arise only in large, long-trained models. Along the way we found that a shared learning rate of 10^{-3} breaks ViT-Base, so the experiment uses $2.5{\times}10^{-4}$ for every size. Finally, we deploy a FastAPI web app that classifies an uploaded image and shows, side by side, how registers clean up DINOv2’s attention and norm maps.

1. Introduction and problem statement

A Vision Transformer (ViT) cuts an image into small square patches, turns each patch into a vector (a *token*), and lets all tokens exchange information through attention. A common way to see what such a model has learned is to look at the attention of its summary token, [CLS], over the patches. For the self-supervised model DINO these maps neatly outline the main object, which made them useful for finding objects without labels. Its successor DINOv2, and also supervised (DeiT-III) and text-supervised (OpenCLIP) ViTs, instead show a few very bright spots on empty background. These spots make the maps hard to interpret and hurt methods that rely on clean feature maps.

The paper asks what these spots are, why they appear, and how to remove them. This matters because ViT features are reused as general-purpose building blocks for many tasks (segmentation, depth estimation, object discovery), and those tasks need every patch token to describe its own part of the image.

Our goals were to (i) understand the paper and its context, (ii) reproduce its central measurements, (iii) implement the method ourselves and train it within a free-GPU budget, (iv) run our own experiment on whether the problem appears at small model sizes, and (v) deploy the result as a working app.

2. Literature and background

Before the paper. Adding special tokens to a transformer is an old idea: BERT and ViT add [CLS] to read out a prediction, and DETR adds object queries. In these cases the extra tokens carry information in or results out. The closest precursor is the *Memory Transformer* (Burtsev et al., 2020), which added blank memory tokens to a language model and improved translation; Sandler et al. (2022) tried learnable memory tokens for fine-tuning ViTs, with limited transfer between tasks. On the vision side, DINO (Caron et al., 2021) showed clean attention maps from self-supervised training, and LOST (Siméoni et al., 2021) used them to localise objects without labels. DINOv2 (Oquab et al., 2024) gave much stronger features, yet LOST worked poorly on it. That puzzle is the paper's starting point. MAE (He et al., 2022), trained only to rebuild hidden patches, shows no such spots.

What was missing. Nobody had explained why most modern ViTs have noisy attention maps while DINO does not, and extra tokens had been treated as an accuracy add-on rather than a fix for a problem inside the model.

After the paper. Sun et al. (2024) found similar "massive activations" in language models and ViTs and argue they act as fixed biases in attention, so registers can be seen as learned biases. Yang et al. (2024) study a different, grid-like artifact caused by position embeddings, which registers do not remove. Jiang et al. (2025) locate a few neurons that create the high-norm tokens and obtain most of the benefit of registers at test time, without retraining. Together these confirm that the phenomenon is real, while refining the explanation.

3. Paper methodology

Diagnosis. The authors look at the length (L2 norm) of every output patch token. In DINOv2 ViT-g the distribution has two humps: most tokens are below about 100, while roughly 2% are far longer. They call tokens above a hand-picked cutoff of 150 *high-norm* or *outlier* tokens and show that they (a) appear after about a third of training, around the middle layers, and only in large models (ViT-L and above); (b) sit on patches that look almost identical to their neighbours, i.e. redundant background; (c) have lost information about their own position and pixels; and (d) contain much more information about the whole image than normal patches.

Explanation. A large, well-trained model learns to recognise patches it does not need and re-uses their tokens as scratch space for global computation. This is not harmful in itself, but doing it inside patch tokens destroys their local information.

Fix. Add N learnable *register* tokens to the input sequence, next to [CLS] and the patches. They carry no image content, receive no position embedding, and are discarded at the output. The model can now use them as scratch space and leaves the patches alone. With 4 registers the extra compute is under 2%. The paper reports that registers remove the outliers for DeiT-III, OpenCLIP and DINOv2, leave ImageNet accuracy essentially unchanged (−0.1 to +0.5 points), improve dense prediction, and raise LOST object discovery on VOC 2007 from 35.3 to 55.4 CorLoc for DINOv2.

4. Implementation details

Our code is organised as one module per stage of the pipeline (data, model, training, analysis, plots, report) and every component is tagged in PROVENANCE.md as written, adapted or reused. Results are written into the repository README by the scripts themselves, so reported

numbers always come from the code.

Part A — the paper's own models. We load Meta's released DINOv2 ViT-S/B/L/g (patch size 14), each trained with and without 4 registers, through torch.hub. A hook captures the input of the last attention layer; we recompute softmax($qk^T/\sqrt{d}$) to get [CLS] attention maps, and checked that this reproduces the layer's own output to within 10^{-7}. Norms are measured on the last block's output.

Part B — our own ViT. We implemented a standard pre-norm ViT in PyTorch with an option for R registers. Following the official DINOv2 code, registers are inserted after the position embeddings (so they have no position), between [CLS] and the patches, start as small random values (std 10^{-6}), and are dropped at the output, where only [CLS] feeds the classifier. For 64×64 images we use 8×8 patches (64 patch tokens). Training follows the DeiT recipe, scaled down: AdamW (peak learning rate 10^{-3} for the runs in Section 6.2 and 2.5×10^{-4} in the experiment, see Section 7; weight decay 0.05), 5 warm-up epochs then cosine decay, label smoothing 0.1, stochastic depth 0.1, gradient clipping, and mixed precision; batch size 256, 50 epochs, seed 0.

Engineering checks. Before any long run we ran 24 automatic checks on a real batch (shapes, labels, normalisation, a starting loss close to ln 200, gradients reaching the registers, and our explicit attention matching PyTorch's fused version), 15 unit tests, and a 5-epoch sanity run on 10% of the data. Training saves its full state after every epoch so a Colab disconnect never loses a run; the tests run automatically on every push.

5. Dataset and preprocessing

The paper trains on ImageNet-22k and LVD-142M (millions of full-size images), which is far beyond a free GPU. We therefore use **Tiny-ImageNet**: 100,000 training and 10,000 validation images of 64×64 pixels in 200 classes, freely downloadable. The official validation set is our test set because test labels are not public, so we always report the last epoch rather than picking the best one. Training images get a random resized crop (35–100% of the image) and a random horizontal flip; all images are converted to RGB and normalised with ImageNet statistics.

For Part A we use 256 validation images of **Imagenette** (10 easy ImageNet classes) at 224×224, because DINOv2 expects larger images than 64×64. **Implication of the substitution:** our absolute accuracies cannot be compared with the paper's ImageNet numbers; only with/without-register differences and the qualitative behaviour can.

6. Reproduction results

6.1 The paper's claims on DINOv2

With the paper's own cutoff of 150, **2.62%** of ViT-g patch tokens are high-norm; the paper reports 2.37%. Table 1 summarises all four sizes using a size-independent rule (norm above 3× the model's median), which for ViT-g gives a cutoff of 149, close to the paper's 150.

Table 1. DINOv2 patch-token norms (last block, before the final LayerNorm), 256 Imagenette images.

Model	Median norm	Max norm (no reg / reg)	% outliers, no reg	% outliers, 4 reg	Neighbour similarity, outlier vs normal
ViT-S/14	20.1	29.7 / 88.6	0.00%	0.00%	n/a
ViT-B/14	53.9	252.2 / 216.8	0.20%	0.00%	0.884 vs 0.608
ViT-L/14	84.7	145.2 / 420.4	0.00%	0.00%	n/a
ViT-g/14	49.6	560.6 / 286.7	2.62%	0.00%	0.871 vs 0.662

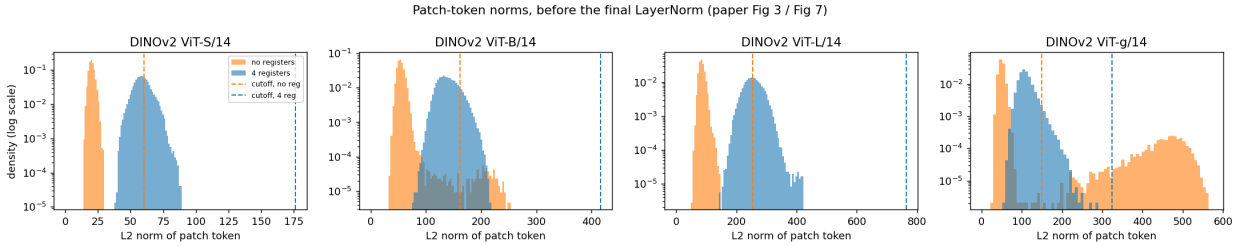


Figure 1. Distribution of patch-token norms (log scale). ViT-g without registers (orange) has a second hump at 300–560; with registers (blue) it is gone. Compare with paper Fig. 3 and 7.

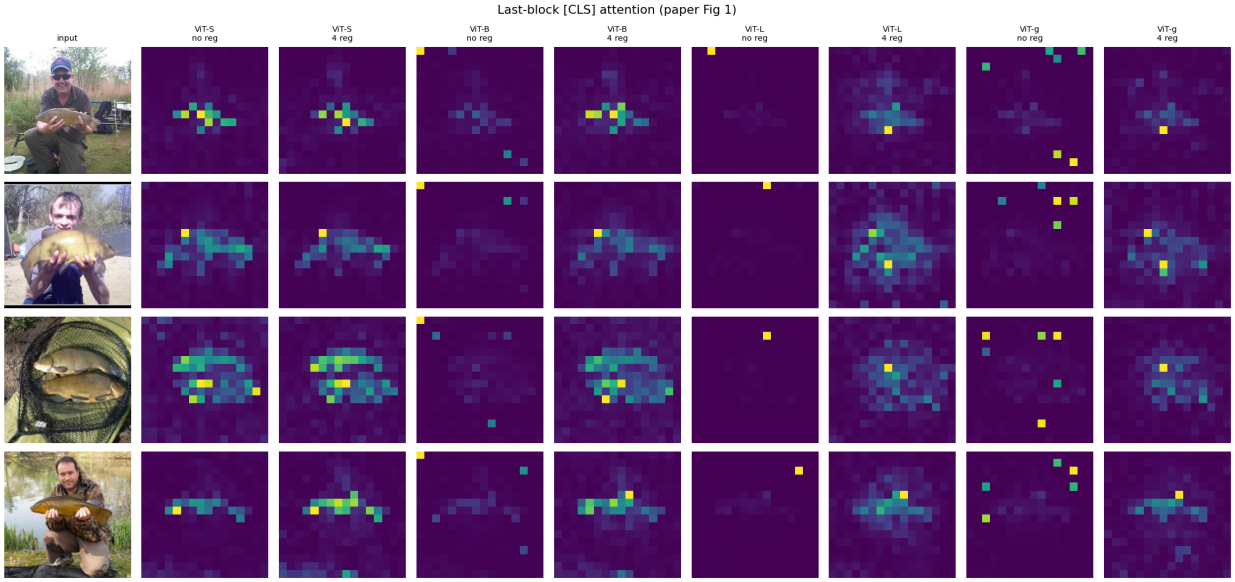


Figure 2. Last-block [CLS] attention. Without registers, ViT-B/L/g show bright isolated spots on the background (often at image edges); with registers attention lies on the object (paper Fig. 1).

- **High-norm outliers exist and are rare** (paper Fig. 3): reproduced, 2.62% vs 2.37%.
- **They sit on redundant patches** (paper Fig. 5a): reproduced. Right after patch embedding, ViT-g outlier patches have a mean cosine similarity of 0.871 to their neighbours versus 0.662 for normal patches, with a sharp peak at 1.0.
- **Registers remove them** (paper Fig. 7) and **clean the attention maps** (paper Fig. 1): reproduced for every size.
- **The large values move into the registers** (paper Fig. 15): reproduced. In ViT-g with registers one register has an average norm of 1390 against 108 for a typical patch, and the four registers behave differently from each other, as in the paper.
- **Outliers appear only in large models** (paper Fig. 4c): partly reproduced. Clear outliers appear only in ViT-g (ViT-B has a tiny tail of 0.20%); the paper also finds them in ViT-L. A likely reason is that Meta's released S/B/L models were distilled from ViT-g rather than trained on their own as in the paper's Fig. 4c.
- **A detail the paper does not stress:** the final LayerNorm rescales every token to a similar length. In ViT-g the largest token is 11.3× the median before it but only 1.08× after it. Our first attempt measured after the LayerNorm and found nothing; the norms must be read before this layer.

6.2 Training our own ViT with and without registers

ViT-Tiny (5.4M parameters) trained for 50 epochs on all of Tiny-ImageNet reaches **38.99%** validation accuracy without registers and **37.86%** with 4 registers (−1.13 points). The training-loss curves lie on top of each other (Figure 3). The paper reports changes of −0.1 to +0.5 points at ImageNet scale, so this claim is only partly reproduced: there is no large

accuracy cost, but a small one cannot be ruled out from a single run per setting, especially since the register run was paused and resumed once (which changes the data order afterwards). Both models overfit (about 91.5% training accuracy; validation loss is lowest around epoch 18), which is typical for ViTs trained from scratch on 100k images and affects both runs equally. In the experiment (Section 7) we retrained ViT-Tiny at a lower learning rate (2.5×10^{-4}): it overfits much less (73% training accuracy) and registers change accuracy by -0.81 points (39.74% vs 38.93%).

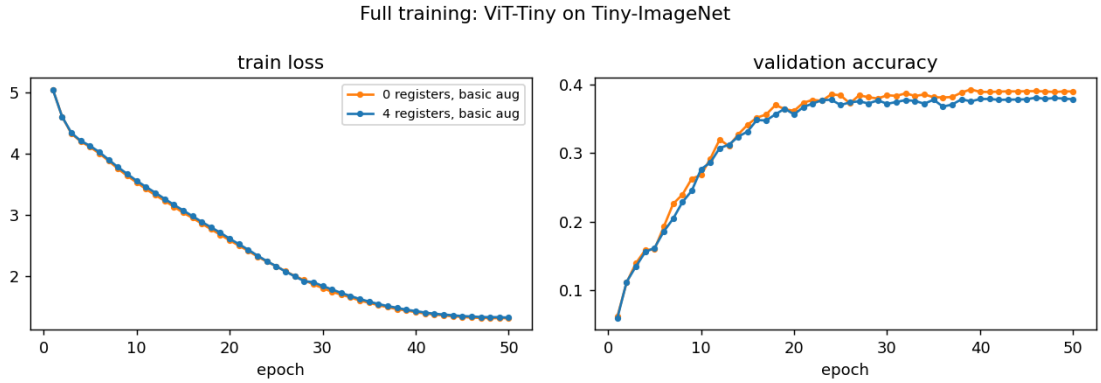


Figure 3. Training loss and validation accuracy of our ViT-Tiny, 0 vs 4 registers, 50 epochs on Tiny-ImageNet.

7. Additional experiment: do artifacts appear with model size?

Question. Do high-norm artifact tokens appear in small ViTs trained from scratch, or only once models are large, and do registers change anything below that size?

Design. A model-size scaling study, the experiment planned in our proposal. We train ViT-Tiny, ViT-Small and ViT-Base, each with 0 and 4 registers, keeping everything else fixed (data, recipe, 50 epochs, seed). We first used the peak learning rate of our ViT-Tiny runs (10^{-3}) for every size. Both ViT-Base models collapsed with it after the warm-up (validation accuracy fell from 12.7% to below 11%) and ViT-Small with registers trained badly, so we retrained all six models at 2.5×10^{-4}, the value DeiT's rule (5×10^{-4} × batch size / 512) gives for our batch size of 256. The 10^{-3} runs are reported separately (Table 3, Figure 8) and are not used in the other figures. For each trained model we measure, on the same 2,000 validation images, the patch-token norms after every block (before the final LayerNorm), the share of outliers (above 3× the median, as in Part A), the largest/median norm, how much attention patches and [CLS] give to the registers (compared with the share they would get if attention were uniform), and [CLS] attention maps. Meta's DINOv2 models from Part A provide the large end of the same size axis. **Hypothesis** from the paper: no outliers in our small models, little use of the registers, and outliers only at the top of the DINOv2 range.

Where to find it. The experiment's code (`scripts/analyse_run.py`, `scripts/summarise_experiment.py`, the measurement functions in `src/analysis.py`), the Colab notebook (`notebooks/experiment_colab.ipynb`), its tests and all 12 runs are in the repository branch `experiment-model-size`.

Table 2. Our trained ViTs (2,000 Tiny-ImageNet validation images, last block, before the final LayerNorm). Uniform attention share in brackets.

ViT	Params	Reg.	LR	Val acc.	Train acc.	Max / median	Outliers	Peak (any block)	Patch attention to registers
Tiny	5.4M	0	2.5e-4	39.74%	73.9%	2.61×	0.00%	3.19×	n/a
Tiny	5.4M	4	2.5e-4	38.93%	73.2%	2.64×	0.00%	3.10×	6.0% (5.8%)
Small	21.5M	0	2.5e-4	39.69%	97.2%	2.04×	0.00%	2.61×	n/a
Small	21.5M	4	2.5e-4	40.13%	97.3%	2.17×	0.00%	2.69×	5.7% (5.8%)
Base	85.4M	0	2.5e-4	40.44%	99.0%	2.32×	0.00%	2.50×	n/a
Base	85.4M	4	2.5e-4	39.78%	99.0%	2.39×	0.00%	2.54×	6.2% (5.8%)

Table 3. First attempt with a peak learning rate of 10^{-3} (not used in the other figures).

ViT	Params	Reg.	LR	Val acc.	Train acc.	Max / median	Outliers	Peak (any block)	Patch attention to registers
Tiny	5.4M	0	1e-3	38.99%	91.6%	2.09×	0.00%	2.31×	n/a
Tiny	5.4M	4	1e-3	37.86%	91.5%	1.84×	0.00%	2.26×	4.1% (5.8%)
Small	21.5M	0	1e-3	33.35%	89.8%	1.67×	0.00%	1.85×	n/a
Small	21.5M	4	1e-3	30.60%	66.5%	1.98×	0.00%	2.36×	1.5% (5.8%)
Base	85.4M	0	1e-3	8.59%	8.9%	1.49×	0.00%	1.64×	n/a
Base	85.4M	4	1e-3	10.27%	10.8%	1.45×	0.00%	1.58×	16.4% (5.8%)

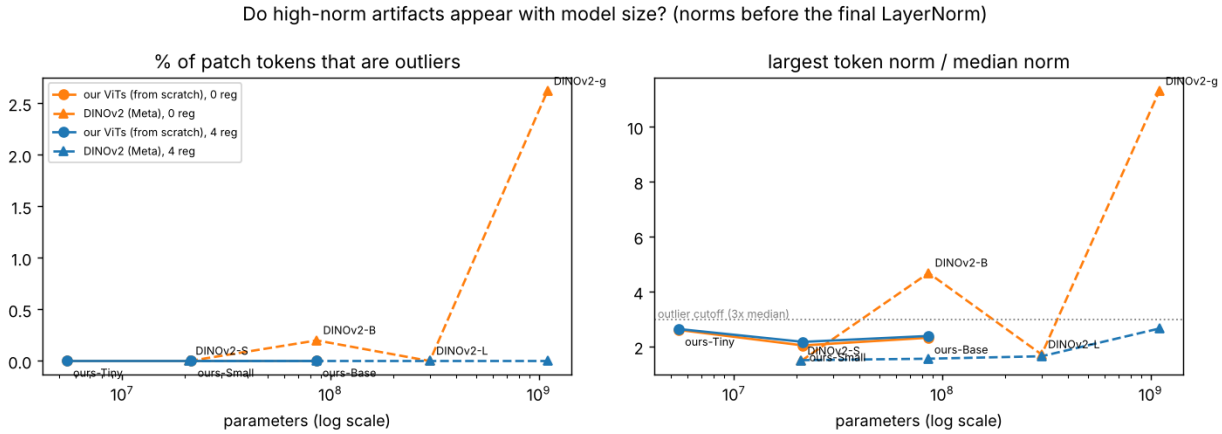


Figure 4. Artifacts against model size: our from-scratch ViTs (circles) and Meta's DINOv2 (triangles), without (orange) and with (blue) registers.

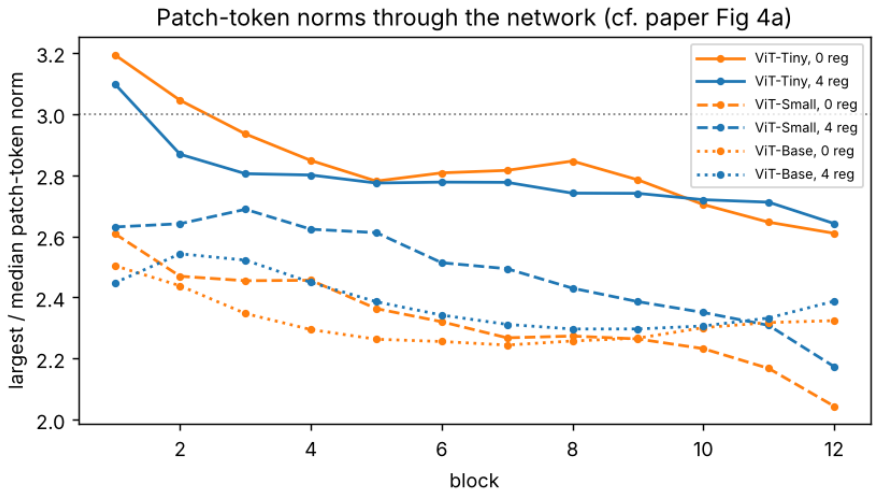


Figure 5. Largest/median patch-token norm after every block of our models (compare with the paper's Fig. 4a).

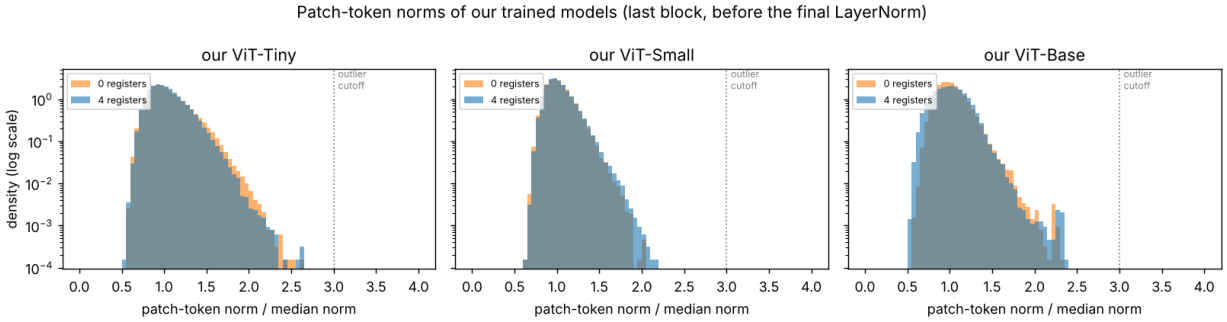


Figure 6. Patch-token norm divided by the model's median norm (last block, before the final LayerNorm). Every model stays far below the 3× outlier line; compare with the second hump of DINOv2 ViT-g in Figure 1.

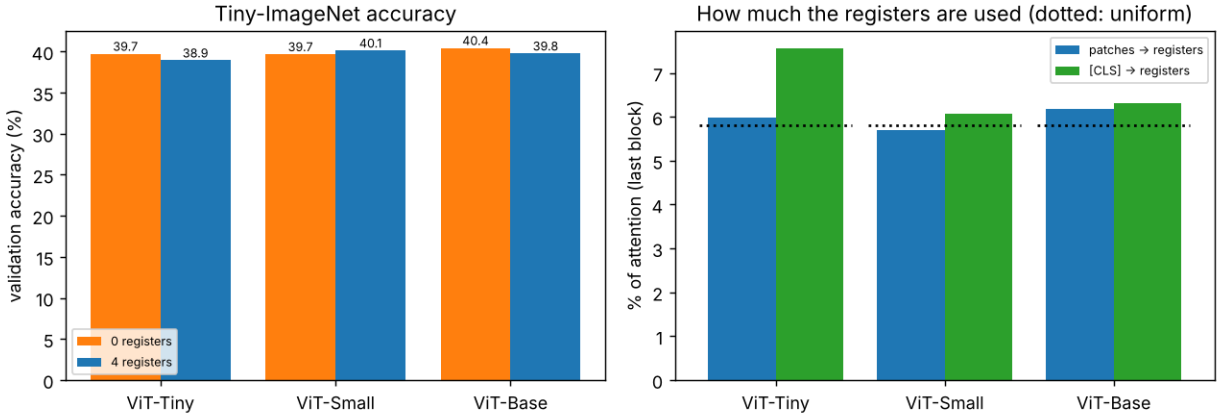


Figure 7. Left: validation accuracy. Right: share of last-block attention that goes to the registers; dotted lines show the share under uniform attention.

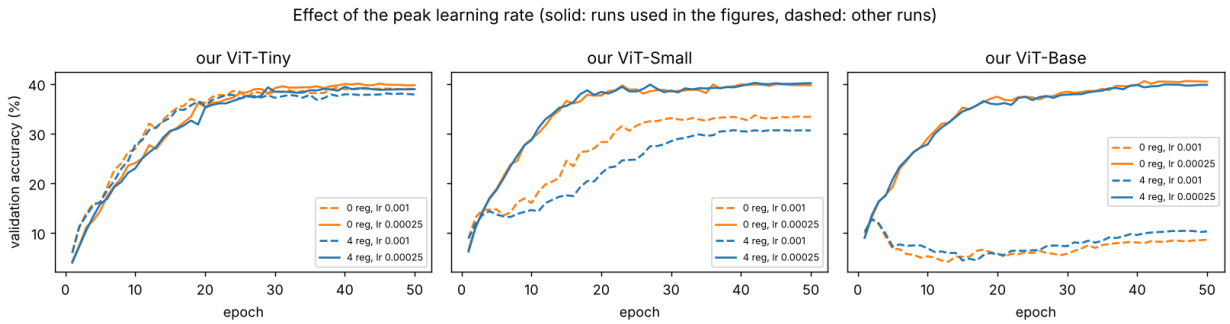


Figure 8. Validation accuracy per epoch at the two peak learning rates. At 10^{-3} ViT-Base collapses after the warm-up and ViT-Small trains worse; at 2.5×10^{-4} all sizes train normally.

8. Analysis and discussion

Do artifacts appear with model size? Not at the sizes we can train. None of our six models has a single outlier token in its last block, where the paper measures them, and the norm histograms have a single hump (Figure 6), unlike the two humps of DINOv2 ViT-g (Figure 1). The only tokens above 3× the median in any block are about 3–6 of 128,000 in the first two blocks of ViT-Tiny, and they are gone by the third block: the opposite of the paper's artifacts, which appear in the middle of the network and persist to the output (paper Fig. 4a). On the shared size axis (Figure 4), clear outliers appear only at the very top, in DINOv2 ViT-g, with a small tail in DINOv2 ViT-B.

Why not? Our ViT-Base has the same size as DINOv2 ViT-B, but it saw 100,000 images of 64×64 pixels for 50 epochs instead of 142 million full-size images. The paper reports that high-norm tokens need both a large model and long training (its Fig. 4b and 4c). Our models are therefore below the point where a model has learned enough to treat background patches as redundant and reuse them. Our experiment changes model size but not training length, so it cannot say which of the two matters more; it does show that size alone, up to 85M parameters, is not enough.

What do registers do in a model without artifacts? Almost nothing. Patch tokens give the four registers 6.0%, 5.7% and 6.2% of their last-block attention in Tiny, Small and Base, and [CLS] gives them 6.1–7.6%; uniform attention would give them 5.8%. Their norms are close to those of a typical patch (ViT-Base: 79–98 against a median patch of 96). In DINOv2 ViT-g with registers, by contrast, one register has a norm of 1390 against 108 for a typical patch. This is the clearest evidence in our project for the paper's explanation: registers are only used as scratch space when the model needs scratch space.

Accuracy. Adding registers changes validation accuracy by −0.81 (Tiny), +0.44 (Small) and −0.66 (Base) points. The paper reports −0.1 to +0.5 points on ImageNet. Our differences are all under a point and have no consistent sign; with one seed per setting we read them as no clear effect, while a small cost cannot be ruled out. Larger models were only marginally more accurate (39.74%, 39.69%, 40.44% without registers) because they memorise the training set (97–99% training accuracy).

The learning rate mattered more than size or registers. With the learning rate of our first ViT-Tiny runs (10^{-3}), ViT-Base collapsed after the warm-up (validation accuracy fell from 12.7% to about 4% and ended below 11%), its token norms grew to about 10,000, and ViT-Small with registers trained so much worse than without registers that the register comparison showed −2.75 points (Table 3, Figure 8). Measuring those models would have produced a misleading result, so we retrained all six models at 2.5×10^{-4}. This is a useful lesson for any size comparison: the training recipe has to work for the largest model before differences between models mean anything.

9. Deployment

We deployed the project as a FastAPI web application (`app/` in the repository). A user uploads an image in the browser, or sends it to the API, and receives:

- a **prediction**: the top-5 Tiny-ImageNet classes from our own trained ViT-Tiny models, without and with registers. The app uses exactly the validation preprocessing from training (resize and centre-crop to 64×64, normalisation), and the weights are exported from our training runs in half precision (11 MB per model);
- **what registers change**: the [CLS] attention map and the patch-token norm map of Meta's DINOv2, without and with registers, side by side, plus the number of outlier patches (norm above 3× the median, the rule used everywhere in this project). DINOv2 is used here because our own small models have no artifacts to show (Section 7). The model size is a setting (`DINOV2_SIZE` = S, B, L or G; default B).

Table 4. The web API.

Endpoint	Input	Output
GET /	—	the upload page (HTML)
POST /predict	multipart form, one image file named *file* (JPEG, PNG, …; up to 10 MB)	JSON: top-5 classes and probabilities per classifier; per DINOv2 model the outlier count, largest/median norm and the two maps as base64 PNG images; processing time
GET /health	—	whether the models are loaded
GET /docs	—	interactive API documentation generated by FastAPI

Engineering. All four models are loaded once when the server starts (FastAPI lifespan), not on every request. Uploads are validated: a file that is not an image returns HTTP 400 and a file over 10 MB returns 413. Attention maps are computed with the same hook-based code as Part A, and norms are read before the final LayerNorm. Both norm maps share one colour scale (norm divided by the model's median, yellow at 3× and above), so the two models can be compared directly; an earlier version scaled each map to its own range, which made small differences look large. Five automated tests start the app with stand-in models and check every endpoint and the error cases; they run on every push. On a laptop CPU a request takes about 1.6–2 s with DINOv2-B and 8 s with DINOv2-L.

Worked example (Figure 9, DINOv2-L). For a photo of sky above grass, DINOv2 without registers puts most of its [CLS] attention on a single patch of empty sky, the artifact of the paper's Fig. 1; with registers the attention moves to the grass, the only content in the image. Neither ViT-L model has a patch above 3× the median (largest/median 1.64× without registers, 1.16× with), which agrees with Part A, where only ViT-g had clear norm outliers. On a photo of a spider web both of our classifiers answer "spider web" (30% without registers, 61% with).

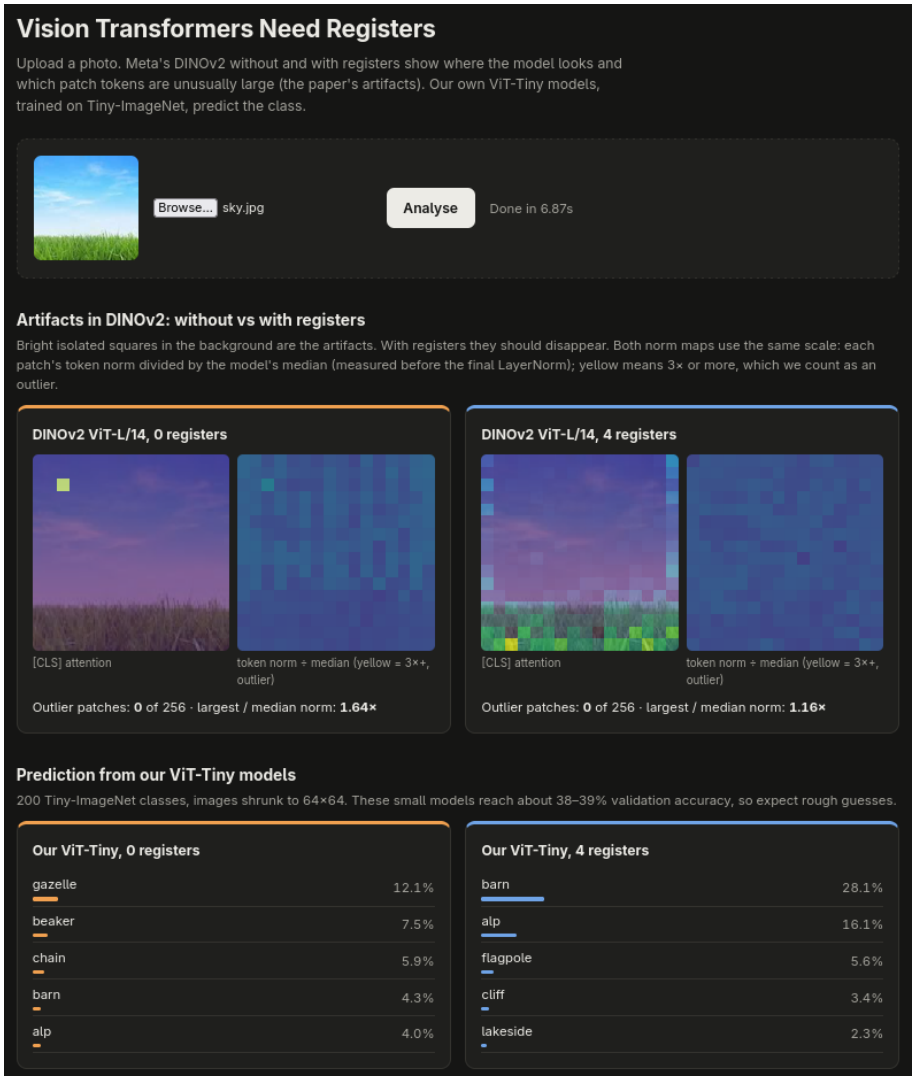


Figure 9. The deployed app on a photo of sky above grass (DINOv2-L): without registers the attention sits on one sky patch; with registers it moves to the grass. Below: our ViT-Tiny's top-5 classes.

10. Limitations

- **Compute:** Google Colab GPUs (T4, L4 and A100 when available). We could not train models of the paper's size or on its data, so large-model results come from Meta's released checkpoints and our own models are small. Some runs moved between GPU types or were paused and resumed from their last saved epoch (ViT-Tiny with registers after epoch 14; the first ViT-Base with registers after epoch 4); resuming changes the order of the training images afterwards.
- **Dataset substitution:** Tiny-ImageNet (64×64) instead of ImageNet-22k/LVD-142M; absolute accuracies are not comparable with the paper.
- **Distilled checkpoints:** Meta's released ViT-S/B/L were distilled from ViT-g, so Part A's size trend is not a clean test of the paper's Fig. 4c.
- **One recipe, one seed:** we did not tune hyper-parameters per size; the learning rate of 2.5×10^{-4} was chosen by DeiT's rule after 10^{-3} failed for ViT-Base. With one seed per setting, accuracy differences of about one point may be noise.
- **Training length:** the paper finds that artifacts appear only after long training (its Fig. 4b). Our 50 epochs on 100,000 images are far shorter than DINOv2's training, so our experiment cannot separate the effect of model size from the effect of training length.
- **Overfitting** of our from-scratch ViTs: ViT-Small and ViT-Base reach 97–99% training accuracy against about 40% on validation (ViT-Tiny: 73%).

- **Not tested:** the probing results on what high-norm tokens contain (Table 1, Fig. 5b of the paper) and the downstream benefits of registers (dense prediction, LOST object discovery; Tables 2–3).
- **Sample sizes:** 256 Imagenette images for Part A and 2,000 validation images for the experiment.
- **Deployment:** the app runs on CPU; DINOv2-L takes about 8 s per image and the giant model, the only one with clear norm outliers, needs about 9 GB of memory. Our classifiers know only 200 classes at 64×64, so predictions on everyday photos are rough.

11. Conclusion

We reproduced the paper's central diagnosis on the models it studied: in DINOv2 ViT-g, 2.62% of patch tokens have a very large norm (paper: 2.37%); they sit on redundant background patches; registers remove them and clean up the attention maps; and the large values move into the registers. A practical detail the paper does not stress is that these norms are only visible before the model's final LayerNorm. The model-size claim is only partly reproduced on the released checkpoints, most likely because Meta's smaller released models were distilled from ViT-g.

Our own ViT with registers, written from scratch, trains normally, and registers change its accuracy by under a point. Our experiment shows that ViTs of 5M–85M parameters trained from scratch on Tiny-ImageNet develop no high-norm artifacts, and that their registers then stay idle, receiving only chance-level attention. This supports the paper's explanation that registers are scratch space a model only needs once it is large and trained long enough. The deployed FastAPI app makes the effect visible on any photo. Next steps would be longer training or more data to find where artifacts first appear, several seeds per setting, and tests of the paper's downstream claims (linear probing, LOST object discovery) on the released models.

12. References

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Provenance and AI use. *Every number and figure in this report was produced by our code (results/ in the repository); numbers attributed to the paper are the authors'. DINOv2 code and weights are Meta's (Apache-2.0), loaded unchanged. PROVENANCE.md tags each code component as written, adapted or reused. Code and text were drafted with the help of an AI assistant (Claude, Anthropic) and reviewed, run and tested by us.*